SECTION 3 · SUBSECTION 3.9
OPERATOR DISPLAY

3.9.1 INTRODUCTION.

The operator display is a single-page React application in *frontend/*. It has one job: submit a repository address, show which of the seven stages is running, then present the returned documents in six tabs. Navigation is a local state variable, not a URL router. The bundle is built by Vite and served as static files; every dynamic request goes to the FastAPI origin named by *VITE_API_URL* (figure 3.9-1).

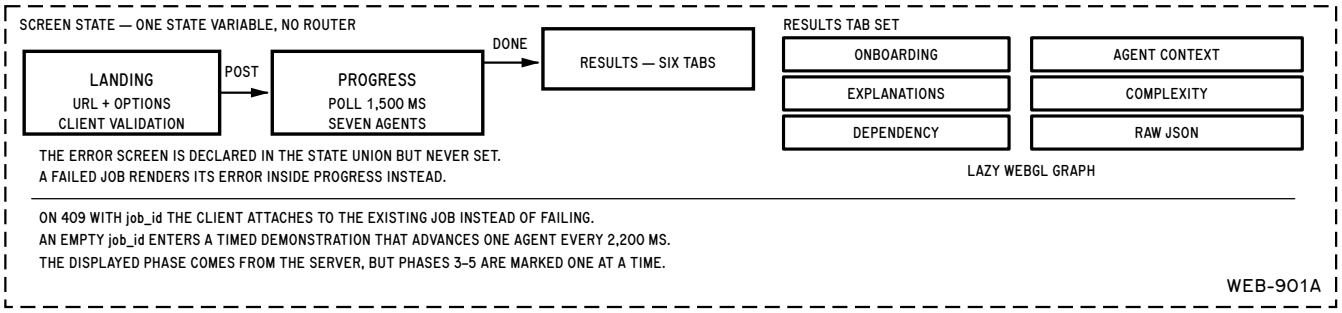


Figure 3.9-1. Client Screen Transitions and Result Tabs

3.9.2 FUNCTIONAL DESCRIPTION.

The landing screen accepts a repository address and two options: an explanation cap and a skip-LLM switch. It rejects anything that does not begin with *http(s)://github.com/owner/repo* before calling the API. A successful submission returns a job identifier; the client switches to the progress screen and requests */api/jobs/{id}* immediately, then every 1,500 ms. A non-OK response is ignored and polling continues, which makes the display tolerant of a restarting server. Terminal status stops the timer; unmounting clears it.

Progress is computed on the client: completed agents plus half credit for the running agent, divided by seven. The agent names and statuses themselves come from the API. When the job completes, the results screen fetches */api/jobs/{id}/result* once and renders the Markdown, the JSON and the graph payload. The 3D graph module is split into a lazy chunk so the first paint does not pay for *three* and *3d-force-graph*.

3.9.2.1 Document Presentation.

Markdown is rendered with a GitHub-flavoured parser and a table extension. Downloads are produced in the browser from the strings already in memory, so the API has no download endpoint. Raw JSON exposes the serialised result for inspection. The dependency tab builds a force-directed graph from nodes and links; node colour is the risk level, and the circular-dependency flag is shown in the side panel.

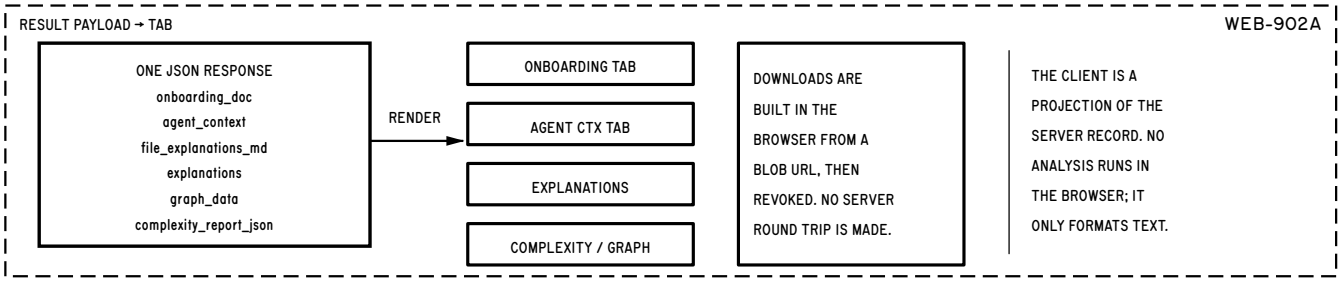


Figure 3.9-2. Single Payload, Client-Side Rendering and Download

3.9.3 PERFORMANCE AND DESIGN DATA.

Table 3.9-1. Client Runtime Constants

Item	Value	Source
API origin	VITE_API_URL or same origin	App.tsx
Poll interval	1,500 ms	App.tsx
Demo tick	2,200 ms	App.tsx
Agents displayed	7, server supplied	/api/jobs/{id}
Result tabs	6	App.tsx
App version string	1.1.0	App.tsx
3D graph bundle	lazy chunk	React.lazy
Router dependency	installed, unused	package.json

CAUTION

The client trusts the server's agent list to describe concurrency. The API marks graph, complexity and retrieval as separate sequential phases even while their futures overlap, so the display understates parallelism. The progress percentage is therefore a presentation estimate, not a measured completion fraction.

DISPLAY CHECK	OPERATOR
1 SUBMIT - bad URL rejected	
2 SUBMIT - duplicate job attaches via 409	
√ POLL - seven agent rows update	
3 OPEN - all six result tabs	
4 VERIFY - downloads open offline	